RESEARCH WRITING SAMPLE

Methodology & Results

Selected excerpts from the final MSc dissertation

About this sample

This document contains selected pages and sections reproduced directly from my final 89-page MSc dissertation officially submitted to Trinity College Dublin. The dissertation pages that follow are copied from the submitted PDF without rewriting or altering their equations, tables, figures, or reported results. This added page introduces the selection; it is not part of the dissertation.

The excerpts cover the logistic-regression and neural-network methods, the complete primary test-results table, overall and disagreement-match comparisons, uncertainty analysis, and a selected bibliography page. The included bibliography page is an excerpt, not the complete dissertation bibliography.

Source dissertation

When Ratings and Markets Disagree: TTT-Informed Covariate Models for Tennis Match Prediction. Zhiheng Huang. MSc in Statistics and Sustainability, University of Dublin, Trinity College, 2026.

Page selection

Original printed dissertation pages 24 and 26; 42–49; and selected bibliography page 64. Page numbers printed on the reproduced pages refer to the full dissertation. This sample is dissertation work, not a peer-reviewed publication.

$$\log\left(\frac{p_i^{LR}}{1-p_i^{LR}}\right) = \eta_i.$$

This formulation allows the intercept and the coefficient attached to the TTT prediction to correct systematic bias or overconfidence in the TTT baseline probabilities. The remaining coefficients then determine whether the additional covariates provide predictive information beyond the rating-based estimate.

Eight candidate specifications were fitted, as presented in Table 3.3. The first models examined TTT recalibration and the separate contributions of ranking and historical form. These information groups were then combined in the full covariate model. Short-term rest and workload measures were subsequently added, followed by direct head-to-head and common opponent information, considered separately and jointly.

Table 3.3: Logistic regression candidate specifications

Models	Information
TTT calibration	Log-odds transformation of the original TTT probability
TTT + ranking	TTT probability and uncertainty, rankings, ranking points, and absolute strength gaps
TTT + form	TTT information, career and recent form, surface records, experience, and rest-related information
Full covariates	Combined ranking and form information, together with surface, tournament series, and round
Full + short-term	Full covariate model with additional short-term rest and workload variables
Full + short-term + head-to-head	Short-term model with direct head-to-head information
Full + short-term + common opponent	Short-term model with common opponent information
Full + short-term + head-to-head + common opponent	Short-term model with both relational feature groups

The TTT outputs were used as predictors in a separate logistic regression rather than incorporated into the internal TTT rating process. Bookmaker probabilities were excluded from every specification and retained solely as an external benchmark. Meanwhile, a complete list of variables for each model will be included in the Appendix.

the network can represent relationships and interactions that do not need to be specified individually in the model formula.

The activations from the hidden layer were then combined in a single output unit. The output layer linear predictor was defined as

$$\eta_i^{NN} = b_o + \sum_{j=1}^{H} a_j \, h_{ij},$$

where a_j is the weight connecting hidden unit j to the output unit and b_o is the output bias. A logistic output function converted this value into the predicted probability that Player A wins match i:

$$p_i^{NN} = \frac{1}{1 + exp\left(-\eta_i^{NN}\right)}.$$

The resulting prediction is bounded between zero and one and is therefore directly comparable with the TTT baseline and bookmaker probabilities.

The network parameters were estimated using entropy fitting, corresponding to binary cross-entropy for the observed match outcome. Weight decay was included to discourage excessively large network weights and reduce the risk of fitting highly irregular patterns in the training data. The penalized objective can be represented as

$$\mathcal{J}(\theta) = -\sum_{i=1}^{n_{\text{train}}} \left[Y_i \log\left(p_i^{NN}\right) + (1 - Y_i) \log\left(1 - p_i^{NN}\right)\right] + \lambda \sum_{k=1}^{K} w_k^2,$$

where θ denotes the fitted network parameters, θ_j denotes a penalized network weight, and λ is the weight decay parameter. The first component penalizes probability estimates that assign low probability to the observed result, while the second restricts the overall magnitude of the fitted weights. Network optimization used the Broyden-Fletcher-Goldfarb-Shanno (BFGS) algorithm. The initial comparison of the four feature families used a maximum of 300 iterations. Because the selected short-term family showed nonconvergence at this limit, its complete 48-configuration validation grid and all 876 rolling test fits were rerun with a maximum of 1,000 iterations for the final primary analysis. All final short-term validation and test fits converged.

Four neural network families were fitted, as listed in Table 3.4. The short-term specification formed the common base and included the TTT outputs, the main pre-match covariates, and the short-term workload and rest variables. The remaining

Model	Log loss	Brier score	Accuracy
Short-term + head-to-head + common opponent	0.6050	0.2094	0.6570

The short-term specification recorded the lowest validation log loss of 0.6038. It also had the lowest Brier score, at 0.2090, and the highest classification accuracy, at 0.6621. It was therefore selected as the primary neural network model. The combined head-to-head and common-opponent specification recorded the second-lowest validation log loss of 0.6050. The common-opponent and head-to-head specifications produced validation log losses of 0.6071 and 0.6078, respectively.

4.2 Overall Test Performance

4.2.1 Primary Models and Benchmarks

Following the validation-stage selection, the primary logistic regression and neural network models were evaluated on the test set alongside the TTT baseline and bookmaker benchmark. Table 4.3 reports their performance across the 7,725 matches played between 2023 and 2025.

Table 4.3: Test performance of the primary models and benchmarks over 2023-2025

Forecast	Specification	Logloss	Brierscore	Accuracy	Meanprobability
Bookmaker benchmark	—	0.5866	0.2019	68.13%	0.5927
TTT baseline	—	0.6576	0.2275	63.70%	0.5753
Primary logistic regression	Full + short-term + head-to-head	0.6169	0.2148	65.48%	0.5768
Primary neural network	Short-term	0.6159	0.2143	64.96%	0.5783

The bookmaker benchmark recorded the lowest test log loss of 0.5866 and the lowest Brier score of 0.2019. It also produced the highest classification accuracy, at 68.13%, and the highest mean probability assigned to the actual winner, at 0.5927.

The TTT baseline recorded a log loss of 0.6576, a Brier score of 0.2275, and an accuracy of 63.70%. Both primary covariate models recorded lower log loss and Brier score values than the TTT baseline, together with higher classification accuracy. The primary neural network recorded a log loss of 0.6159, a Brier score of 0.2143, and an

accuracy of 64.96%. The primary logistic regression model recorded corresponding values of 0.6169, 0.2148, and 65.48%. The neural network therefore had slightly lower probability error, while logistic regression had slightly higher accuracy. Their mean probabilities assigned to the actual winner were 0.5783 and 0.5768, respectively.

4.2.2 Other Predefined Specifications

The remaining predefined logistic regression and neural network specifications were also evaluated on the same test set. Table 4.4 reports their test performance.

Table 4.4: Test performance of the other predefined specifications

Model	Log loss	Brier score	Accuracy	Mean probability
Logistic: full + short-term	0.6175	0.2151	65.31%	0.5765
Logistic: full + short-term + common opponent	0.6176	0.2151	65.39%	0.5765
Logistic: full + short-term + head-to-head + common opponent	0.6170	0.2148	65.28%	0.5768
Neural network: short-term + head-to-head	0.6168	0.2145	65.63%	0.5783
Neural network: short-term + common opponent	0.6143	0.2138	65.17%	0.5791
Neural network: short-term + head-to-head + common opponent	0.6166	0.2149	65.10%	0.5786

The three logistic regression specifications produced similar test results. Their log loss ranged from 0.6170 to 0.6176, while their classification accuracy ranged from 65.28% to 65.39%. The combined head-to-head and common-opponent specification recorded the lowest log loss among these three models, at 0.6170.

Among the neural network extensions, the common-opponent specification recorded the lowest observed test log loss, at 0.6143, and the lowest Brier score, at 0.2138. The head-to-head specification had the highest classification accuracy, at 65.63%, while the head-to-head and combined specifications recorded log losses of 0.6168 and 0.6166, respectively. Although the common-opponent extension had a lower observed test log loss than the primary short-term neural network, it remained an extension because the primary specification had been selected using validation performance.

4.3 Log-loss Gap Reduction

The test-set log-loss difference between the TTT baseline and the bookmaker benchmark was 0.0709, based on log-loss values of 0.6576 and 0.5866, respectively. Figure 4.1 presents the proportion of this gap closed by the short-term logistic regression and neural network specifications and their predefined extensions.

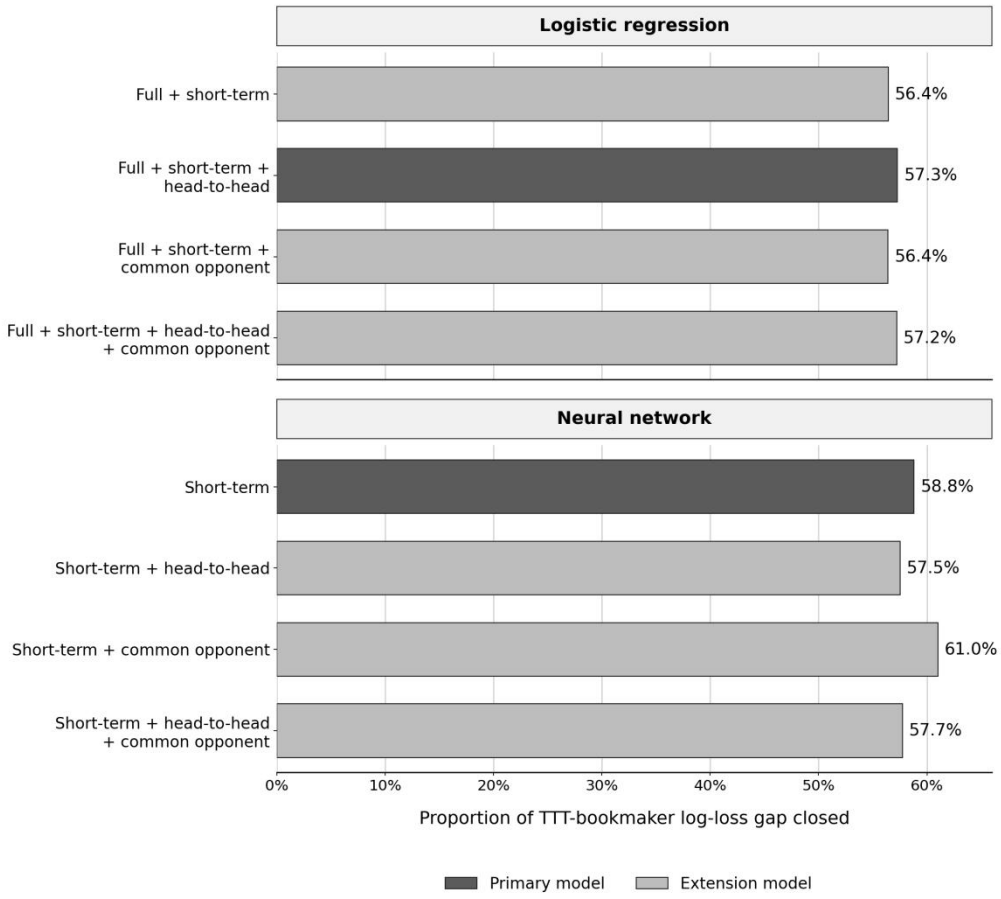


Figure 4.1: Proportion of the test set TTT − bookmaker log loss gap closed by the covariate models. Dark bars indicate the primary models selected by validation performance.

The primary logistic regression model closed 57.3% of the test-set log-loss gap, while the primary neural network closed 58.8%. The difference between the two primary models was 1.5 percentage points. Across the logistic regression specifications, gap reduction ranged from 56.4% to 57.3%. The neural network specifications ranged from 57.5% to 61.0%. Among the predefined extensions, the common-opponent neural network recorded the highest gap reduction, at 61.0%. The primary model designations remained those determined from validation performance.

4.4 Direction Disagreement

4.4.1 Frequency

A direction disagreement occurred when the TTT baseline and bookmaker benchmark assigned probabilities above 0.5 to different players. Table 4.5 reports the number and proportion of disagreement matches in the full sample and each chronological partition.

Table 4.5: Frequency of TTT–bookmaker direction disagreement

Sample	Disagreement matches	Total matches	Disagreement rate
Full sample	7,145	39,154	18.25%
Training set	5,208	28,890	18.03%
Validation set	441	2,539	17.37%
Test set	1,496	7,725	19.37%

Across the full analytical sample, the TTT baseline and bookmaker benchmark predicted different winners in 7,145 of 39,154 matches, corresponding to a disagreement rate of 18.25%. The disagreement rate was 18.03% in the training set and 17.37% in the validation set. In the test set, 1,496 of 7,725 matches produced a direction disagreement, giving the highest partition-specific rate of 19.37%.

4.4.2 Test-set Disagreement Outcomes

Among the 1,496 test matches in which the two forecasts disagreed, exactly one of the bookmaker and TTT baseline directions was correct. Table 4.6 reports the corresponding outcomes.

Table 4.6: Outcomes of test matches that TTT and the bookmaker predicted differently

Outcome	Matches	Proportion
Bookmaker correct, TTT baseline incorrect	919	61.43%
TTT baseline correct, bookmaker incorrect	577	38.57%
Total	1,496	100.00%

The bookmaker benchmark predicted the correct winner in 919 disagreement matches, representing 61.43% of the test disagreement subset. The TTT baseline predicted the correct winner in the remaining 577 matches, corresponding to 38.57%.

4.5 Performance in Disagreement Matches

The predictive metrics were recalculated for the 1,496 test matches in which the TTT baseline and bookmaker benchmark predicted different winners. Table 4.7 compares the two benchmarks, the primary models, and the predefined logistic regression and neural network extensions within this subset.

Table 4.7: Predictive performance within the test-set disagreement subset

Model	Log loss	Brier score	Accuracy	Mean probability
Bookmaker benchmark	0.6553	0.2318	61.43%	0.5345
TTT baseline	0.8239	0.3031	38.57%	0.4683
Logistic: full + short-term	0.7088	0.2571	51.34%	0.5048
Logistic: full + short-term + head-to-head (primary)	0.7065	0.2560	52.01%	0.5058
Logistic: full + short-term + head-to-head + common opponent	0.7060	0.2558	51.40%	0.5061
Neural network: short-term (primary)	0.6973	0.2517	51.20%	0.5118
Neural network: short-term + head-to-head	0.7015	0.2538	52.34%	0.5105
Neural network: short-term + common opponent	0.6995	0.2530	50.94%	0.5115
Neural network: short-term + head-to-head + common opponent	0.7023	0.2540	52.34%	0.5094

The bookmaker benchmark recorded the lowest log loss of 0.6553 and the lowest Brier score of 0.2318. Its classification accuracy was 61.43%, compared with 38.57% for the TTT baseline. The TTT baseline also recorded the highest log loss, at 0.8239, and the highest Brier score, at 0.3031.

The primary logistic regression model reduced log loss to 0.7065, while the primary neural network recorded a lower value of 0.6973. Their Brier scores were 0.2560 and 0.2517, and their accuracies were 52.01% and 51.20%, respectively. The primary neural network recorded the lowest log loss and Brier score among the fitted covariate models. The head-to-head and combined neural network extensions recorded the highest accuracy, both at 52.34%.

The log-loss gap between the TTT baseline and bookmaker benchmark within the disagreement subset was 0.1686. The primary logistic regression model closed 69.6% of

this gap, while the primary neural network closed 75.1%. Among the neural network extensions, the common-opponent specification closed 73.7%, the head-to-head specification closed 72.6%, and the combined specification closed 72.1%. The primary neural network therefore recorded the largest gap reduction among the fitted covariate models in this subset.

4.6 TTT Uncertainty

4.6.1 Disagreement Rate

Figure 4.2 presents the proportion of test matches producing a TTT–bookmaker direction disagreement within each TTT uncertainty group. The group boundaries were derived from the training-set uncertainty distribution, as described in Section 3.4.3.

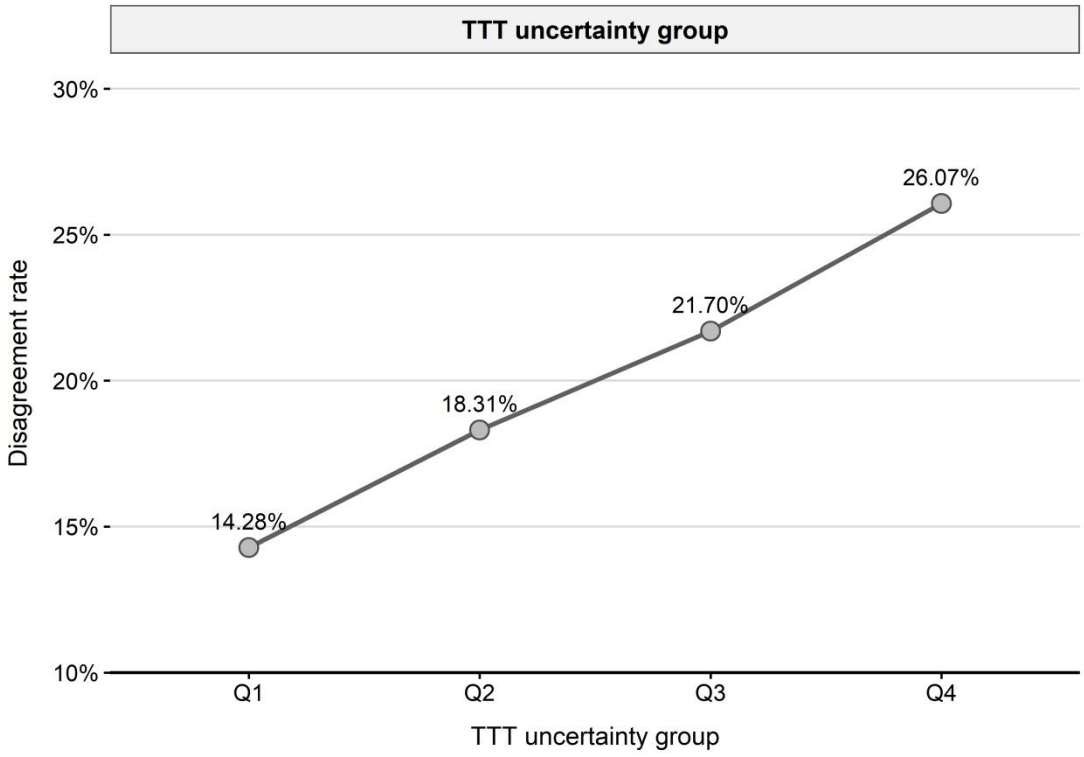


Figure 4.2: TTT – bookmaker disagreement rate across TTT uncertainty groups in the test set. Q1 represents the lowest uncertainty and Q4 the highest.

The disagreement rate increased across the four uncertainty groups. It was 14.28% in Q1, rising to 18.31% in Q2 and 21.70% in Q3. The highest rate was recorded in Q4, where 26.07% of test matches produced different predicted winners from the TTT baseline and bookmaker benchmark. The difference between Q1 and Q4 was 11.79

percentage points.

4.6.2 Prediction Error

To examine how the overall performance differences reported in Section 4.5 varied across uncertainty levels, Figure 4.3 presents log loss for the bookmaker benchmark, TTT baseline, primary logistic regression model, and primary neural network model across the four TTT uncertainty groups. Q1 represents the lowest uncertainty and Q4 the highest. The groups contained 332, 396, 377, and 391 disagreement matches, respectively.

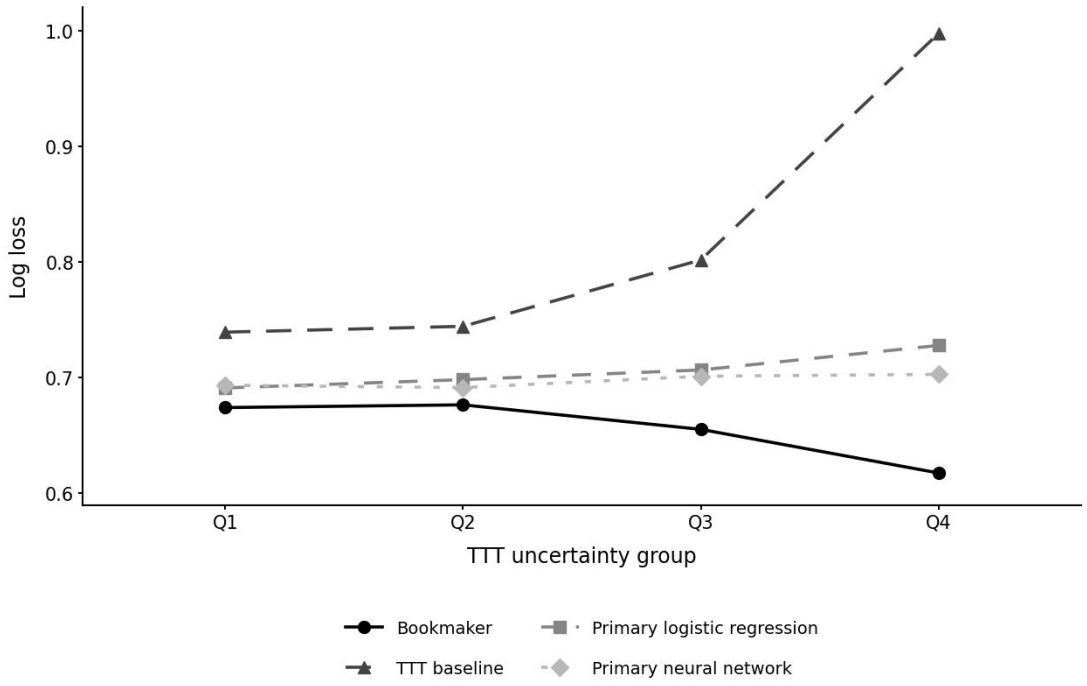


Figure 4.3: Log loss across TTT uncertainty groups within test set disagreement matches. Q1 represents the lowest uncertainty and Q4 the highest; lower log loss indicates better predictive performance.

The TTT baseline recorded similar log loss in Q1 and Q2, at 0.7393 and 0.7444. Its log loss then increased to 0.8017 in Q3 and 0.9975 in Q4. By comparison, bookmaker log loss was 0.6742 in Q1, 0.6765 in Q2, 0.6553 in Q3, and 0.6176 in Q4. The difference between the two benchmarks therefore increased from 0.0651 in Q1 to 0.3799 in Q4.

Both primary fitted models recorded lower log loss than the TTT baseline in every group. Logistic regression log loss was 0.6911 in Q1, 0.6983 in Q2, 0.7067 in Q3, and 0.7278 in Q4. The neural network recorded corresponding values of 0.6934, 0.6914,

0.7012, and 0.7029. Logistic regression recorded the lower value in Q1, whereas the neural network recorded the lower value in Q2, Q3, and Q4.

The differences between the two fitted models were small in Q1 to Q3 and largest in Q4. Their separation from the TTT baseline was also greatest in Q4, where log loss was 0.7278 for logistic regression and 0.7029 for the neural network, compared with 0.9975 for TTT.

4.7 Summary of Findings

Based on the results obtained, it is clear to see that the full short-term logistic regression model with head-to-head information and the short-term neural network were selected as the two primary specifications. On the test set, both primary models improved on the TTT baseline across log loss, Brier score, and classification accuracy, although neither reached the bookmaker benchmark. In addition, their overall performance was actually very similar, the neural network recorded slightly lower log loss and Brier score, while logistic regression recorded slightly higher accuracy. The predefined extensions also produced similar results, with the common-opponent neural network recording the lowest observed test log loss among the fitted models.

Then the covariate-based models reduced more than half of the overall log-loss gap between TTT and the bookmaker benchmark, and this improvement was larger within the disagreement subset, where TTT and the bookmaker predicted different winners. Such disagreements occurred in approximately one fifth of test matches, and the bookmaker identified the correct winner more often than TTT. Within these matches, both primary models substantially reduced TTT prediction error, and the primary neural network recorded the largest gap reduction among the fitted covariate models.

Subsequently, the uncertainty analysis showed a consistent pattern across the test set that TTT-bookmaker disagreements became more frequent as TTT uncertainty increased. Prediction error for the TTT baseline also rose across the uncertainty groups and was highest in Q4. By contrast, the bookmaker benchmark and the two primary fitted models showed smaller changes across the groups. Both primary models recorded lower log loss than TTT at every uncertainty level, with the largest difference observed in the highest-uncertainty group.

To sum up, the results show that additional pre-match covariates improved the TTT

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